

SpaceKLIP Installation Instructions

Separate macOS/Linux and Windows instructions with required and optional steps. Directory paths are examples and can be changed.

Note: the Python module to import is **spaceKLIP**. Do not use lowercase spaceklip for the verification check.

macOS / Linux

Step 1. Create the environment

Status: Required

Use an isolated conda environment with Python 3.11. The SpaceKLIP docs state that Python 3.12 is not currently compatible.

```
conda create -n spaceklip python=3.11 -y
conda activate spaceklip
python -m pip install --upgrade pip setuptools wheel
```

Step 2. Clone the repository anywhere you prefer

Status: Required

The location is not fixed. The example below uses \$HOME/src/spaceKLIP. Change only SPACEKLIP_REPO if you want a different location.

```
export SPACEKLIP_REPO="$HOME/src/spaceKLIP"
mkdir -p "$(dirname "$SPACEKLIP_REPO")"
git clone https://github.com/spacetelescope/spaceKLIP.git "$SPACEKLIP_REPO"
cd "$SPACEKLIP_REPO"
```

Step 3. Install dependencies and SpaceKLIP

Status: Required

Run these inside the cloned repository, following the SpaceKLIP installation page.

```
pip install -r requirements.txt
pip install -e .
```

Step 4. Configure the CRDS cache

Status: Required unless already configured for the JWST pipeline

This tells the JWST pipeline where to cache calibration reference files. The location is only an example.

```
mkdir -p "$HOME/data/crds_cache"
export CRDS_PATH="$HOME/data/crds_cache"
export CRDS_SERVER_URL="https://jwst-crds.stsci.edu"
```

Step 5. Save the CRDS settings in your shell

Status: Recommended

Add them to the startup file for the shell you use. Use ~/.zshrc for zsh or ~/.bashrc for bash.

```
cat >> ~/.zshrc <<'EOF'
export CRDS_PATH="$HOME/data/crds_cache"
export CRDS_SERVER_URL="https://jwst-crds.stsci.edu"
EOF
source ~/.zshrc
```

Step 6. Decide how to handle STPSF data

Status: Automatic by default; set only if you want a custom location

STPSF normally downloads the required data into `$HOME/data/stpsf-data` automatically. For a custom or offline location, download `stpsf-data-LATEST.tar.gz` from the STPSF page, extract it, and point `STPSF_PATH` to the extracted directory.

```
mkdir -p "$HOME/data"
tar -xzf "$HOME/Downloads/stpsf-data-LATEST.tar.gz" -C "$HOME/data"
export STPSF_PATH="$HOME/data/stpsf-data"
```

Step 7. Configure the webbpsf_ext cache

Status: Optional but recommended

`WEBBPSF_EXT_PATH` is not a reference-file directory. It is the cache for PSF coefficients generated by `webbpsf_ext`.

```
mkdir -p "$HOME/data/webbpsf_ext_data"
export WEBBPSF_EXT_PATH="$HOME/data/webbpsf_ext_data"
```

Step 8. Configure CDBS for synphot/stsynphot

Status: Optional; use it if a calculation requires it

Some `webbpsf_ext` or spectral-response workflows may require CDBS data. Download the CDBS data from the `webbpsf_ext` README, extract it, and point `PYSYN_CDBS` to the extracted directory.

```
mkdir -p "$HOME/data/cdbbs"
tar -xzf "$HOME/Downloads/cdbbs.tar.gz" -C "$HOME/data/cdbbs"
export PYSYN_CDBS="$HOME/data/cdbbs"
```

Step 9. Register a Jupyter Notebook/Lab kernel

Status: Optional but practical

This makes the same conda environment selectable from Notebook or JupyterLab.

```
conda activate spaceclip
pip install ipykernel
python -m ipykernel install --user --name spaceclip --display-name "Python (spaceclip)"
```

Windows PowerShell

Step 1. Create the environment

Status: Required

Run this in Anaconda Prompt or Miniforge Prompt. Use an isolated Python 3.11 environment.

```
conda create -n spaceklip python=3.11 -y
conda activate spaceklip
python -m pip install --upgrade pip setuptools wheel
```

Step 2. Clone the repository anywhere you prefer

Status: Required

The location is not fixed. The example below uses `$HOME\src\spaceKLIP`. Change only `$env:SPACEKLIP_REPO` if you want a different location.

```
$env:SPACEKLIP_REPO = "$HOME\src\spaceKLIP"
New-Item -ItemType Directory -Force (Split-Path $env:SPACEKLIP_REPO)
git clone https://github.com/spacetelescope/spaceKLIP.git $env:SPACEKLIP_REPO
Set-Location $env:SPACEKLIP_REPO
```

Step 3. Install dependencies and SpaceKLIP

Status: Required

Run these inside the cloned repository.

```
pip install -r requirements.txt
pip install -e .
```

Step 4. Configure CRDS for this PowerShell session

Status: Required unless already configured for the JWST pipeline

This tells the JWST pipeline where to cache calibration reference files. The location is only an example.

```
New-Item -ItemType Directory -Force "$HOME\data\crds_cache"
$env:CRDS_PATH = "$HOME\data\crds_cache"
$env:CRDS_SERVER_URL = "https://jwst-crds.stsci.edu"
```

Step 5. Save CRDS as user environment variables

Status: Recommended

This makes the settings available to new PowerShell sessions and Jupyter. Open a new terminal after saving.

```
[Environment]::SetEnvironmentVariable(
    "CRDS_PATH", "$HOME\data\crds_cache", "User")
[Environment]::SetEnvironmentVariable(
    "CRDS_SERVER_URL", "https://jwst-crds.stsci.edu", "User")
```

Step 6. Decide how to handle STPSF data

Status: Automatic by default; set only if you want a custom location

STPSF normally downloads the required data under the user home directory automatically. For a custom or offline location, download `stpsf-data-LATEST.tar.gz` from the STPSF page, extract it, and point `STPSF_PATH` to the extracted directory.

```
New-Item -ItemType Directory -Force "$HOME\data"
tar -xzf "$HOME\Downloads\stpsf-data-LATEST.tar.gz" -C "$HOME\data"
$env:STPSF_PATH = "$HOME\data\stpsf-data"
[Environment]::SetEnvironmentVariable(
    "STPSF_PATH", "$HOME\data\stpsf-data", "User")
```

Step 7. Configure the webbpsf_ext cache

Status: Optional but recommended

WEBBPSF_EXT_PATH is not a reference-file directory. It is the cache for PSF coefficients generated by webbpsf_ext.

```
New-Item -ItemType Directory -Force "$HOME\data\webbpsf_ext_data"
$env:WEBBPSF_EXT_PATH = "$HOME\data\webbpsf_ext_data"
[Environment]::SetEnvironmentVariable(
    "WEBBPSF_EXT_PATH", "$HOME\data\webbpsf_ext_data", "User")
```

Step 8. Configure CDBS for synphot/stsynphot

Status: Optional; use it if a calculation requires it

Some webbpsf_ext or spectral-response workflows may require CDBS data. Download the CDBS data from the webbpsf_ext README, extract it, and point PYSYN_CDBS to the extracted directory.

```
New-Item -ItemType Directory -Force "$HOME\data\cdb"
tar -xzf "$HOME\Downloads\cdb.tar.gz" -C "$HOME\data\cdb"
$env:PYSYN_CDBS = "$HOME\data\cdb"
[Environment]::SetEnvironmentVariable(
    "PYSYN_CDBS", "$HOME\data\cdb", "User")
```

Step 9. Register a Jupyter Notebook/Lab kernel

Status: Optional but practical

This makes the same conda environment selectable from Notebook or JupyterLab.

```
conda activate spaceclip
pip install ipykernel
python -m ipykernel install --user --name spaceclip --display-name "Python (spaceclip)"
```

Verification: command line

Run this in Terminal on macOS/Linux, or PowerShell/Anaconda Prompt on Windows.

```
conda activate spaceklip
python -c "import importlib.metadata as md; import spaceKLIP, stpsf, webbpsf_ext, jwst, synphot, stsynphot, crds;
print('spaceKLIP', md.version('spaceKLIP')); print('CRDS OK')"
```

Verification: Jupyter Notebook / JupyterLab

Select Python (spaceklip) from the Kernel menu, then run this Python cell.

```
import os
import importlib.metadata as md
import spaceKLIP
import stpsf
import webbpsf_ext
import jwst
import synphot
import stsynphot
import crds

print("spaceKLIP", md.version("spaceKLIP"))
print("CRDS_PATH", os.environ.get("CRDS_PATH"))
print("CRDS_SERVER_URL", os.environ.get("CRDS_SERVER_URL"))
print("STPSF_PATH", os.environ.get("STPSF_PATH"))
print("WEBBPSF_EXT_PATH", os.environ.get("WEBBPSF_EXT_PATH"))
print("PYSYN_CDBS", os.environ.get("PYSYN_CDBS"))
```

Temporary environment variables in Notebook/Lab

Use this Python cell only for the current Notebook/Lab session if persistent variables are not visible yet.

```
import os
from pathlib import Path

home = Path.home()
os.environ["CRDS_PATH"] = str(home / "data" / "crds_cache")
os.environ["CRDS_SERVER_URL"] = "https://jwst-crds.stsci.edu"
os.environ["STPSF_PATH"] = str(home / "data" / "stpsf-data")
os.environ["WEBBPSF_EXT_PATH"] = str(home / "data" / "webbpsf_ext_data")
os.environ["PYSYN_CDBS"] = str(home / "data" / "cdbbs")
```

Primary sources

SpaceKLIP installation

<https://spaceklip.readthedocs.io/en/latest/Installation-and-dependencies.html>

SpaceKLIP GitHub repository

<https://github.com/spacetelescope/spaceKLIP>

spaceKLIP pyproject.toml

<https://raw.githubusercontent.com/spacetelescope/spaceKLIP/develop/pyproject.toml>

spaceKLIP requirements.txt

<https://raw.githubusercontent.com/spacetelescope/spaceKLIP/develop/requirements.txt>

STPSF requirements and data files

<https://stpsf.readthedocs.io/en/latest/installation.html>

webbpsf_ext README

https://github.com/JarronL/webbpsf_ext

JWST Pipeline CRDS documentation

https://jwst-pipeline.readthedocs.io/en/stable/jwst/user_documentation/reference_files_crds.html

JWST Pipeline Python interface

https://jwst-pipeline.readthedocs.io/en/stable/jwst/user_documentation/running_pipeline_python.html

IPython kernels for different environments

https://ipython.readthedocs.io/en/stable/install/kernel_install.html